

**UNIVERSITY OF RUHUNA FACULTY OF SCIENCE
DEPARTMENT OF COMPUTER SCIENCE**



INDUSTRIAL TRAINING

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Reg.No :SC/2022/12901

Date :2026/09/04

Bachelor of Computer Science (General) Degree- 2026

DECLARATION

I hereby declare that this report is based on the government project work carried out by me during the industrial training period. The work presented in this report, except where references are specifically acknowledged, is my own work and has not been submitted previously for another academic qualification. The information presented is correct to the best of my knowledge.

Signature :

Name : Sunitha Sivakumaran

Reg.No : SC/2022/12901

Date : 2026/09/04

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the successful completion of my government project, Welfare Inventory and Distribution Management System (WIDMS). I extend my sincere appreciation to Mr. S.A.S. Lorensuhewa, Industrial Training Coordinator of the Department of Computer Science, University of Ruhuna, for his valuable coordination, encouragement and support throughout the industrial training period. I am especially grateful to my academic supervisor, Ms. Binuiri Raigamkorale, for her continuous guidance, constructive feedback and academic supervision during the requirements analysis, system design, implementation and documentation stages.

I am deeply grateful to Mrs. Rasika U. I. Dissanayaka, Commissioner of the Department of Social Welfare, Probation and Child Care Services - Southern Province, for providing the institutional support and opportunity required to carry out this project. I would also like to express my heartfelt gratitude to my responsible officer, Mr. M.L.A. Sanjeewa, Assistant Commissioner, for his continuous guidance, cooperation and valuable contribution during the requirements gathering and system development process. His explanations of the existing welfare inventory, beneficiary assistance and distribution procedures helped me understand the practical problems faced by the department and develop the system according to actual organizational requirements.

I would also like to thank the officers and staff of the Department of Social Welfare, Probation and Child Care Services - Southern Province, located at the District Secretariat, Galle, for their cooperation, practical knowledge and assistance throughout the project. Their support helped me understand the existing welfare distribution procedures and identify suitable solutions for inventory management, beneficiary tracking, approval workflows and welfare-item distribution.

Furthermore, I sincerely acknowledge the Department of Computer Science, Faculty of Science, University of Ruhuna, and all academic staff members for providing the knowledge, technical guidance and resources required to successfully complete this project. Finally, I thank everyone who directly or indirectly contributed to the successful completion of WIDMS. The knowledge, experience and practical exposure gained through this project have made a valuable contribution to my academic and professional development.

ABSTRACT

The project was proposed to improve the management and distribution of welfare and assistive devices by providing a centralized web-based system for beneficiary management, aid requests, approvals, inventory control, procurement, goods release, distribution, returns, and monitoring. The final system provides role-based access for the **Administrator, Subject Officer, Store Keeper, and Social Service Officer**, together with specialized features for **Vision Camp** and **Contact Lens** processes.

A major challenge during the training was working in a **completely new government administrative environment**. It was necessary to understand the existing welfare procedures, officer responsibilities, approval levels, beneficiary eligibility rules, and item distribution processes before implementing the system. This was addressed through requirement-gathering discussions, continuous communication with responsible officers, process analysis, and repeated validation of the identified requirements.

Another significant challenge was the **frequent change of requirements during development**. As the actual workflow became clearer, several changes were required in user roles, approval stages, Aid Request and Goods Request handling, eligibility rules, inventory and payment procedures, and special workflows. These changes required modifications to the database structure, user interfaces, and system logic. An iterative development approach was followed to review requirements, update affected modules, and retest the complete workflow after each major change.

The system was developed using **HTML, CSS, JavaScript, Bootstrap, PHP, and MySQL/MariaDB**, with XAMPP used as the local development environment. The project provided valuable experience in requirements engineering, database design, web development, debugging, testing, stakeholder communication, and adapting software to changing real-world requirements.

Overall, the project demonstrated the importance of understanding practical organizational processes, responding effectively to changing requirements, and applying continuous testing and stakeholder feedback to develop a system that is more efficient, transparent, and suitable for the actual welfare distribution process

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1 COMPANY PROFILE

1.1 General Introduction



Figure 2: Southern province flag



Figure 1: logo of the organization

Company Name : Department of Social Welfare, Probation & Child Care Services

Address : 5th Floor, District Secretariat Building, Galle, Sri Lanka

Supervisor's Name : Mr. M. L. A. Sanjeewa

Supervisor's Position: Assistant Commissioner

Start Date : 2026/05/25

End Date : 2026/08/30

Contact number : + 94 912 247 836, + 94 912 232 968

Email address : probationspgalle@gmail.com



Figure 3: Emblem of Sri Lanka

The **Department of Social Welfare, Probation & Child Care Services – Southern Province** plays an important role in improving the welfare of vulnerable communities in the Southern Province. The Department provides services for **elderly persons, persons with disabilities, needy and low-income families, vulnerable children, and other individuals who require social assistance**. Its activities include welfare assistance, rehabilitation, child protection, social inclusion, housing and self-employment support, and other programs intended to improve the living conditions of disadvantaged communities.

The history of Probation and Childcare Services in Sri Lanka dates back to **October 1956**, when the service was first established. With the implementation of the **13th Amendment to the Constitution in 1988** and the introduction of Provincial Council administration, probation and childcare services became decentralized and began operating under provincial-level administration. Since then, the Southern Province service has functioned under a **Provincial**

Commissioner, with responsibilities for probation, child care, rehabilitation, and social welfare activities across the province.

Over time, the Department has expanded its responsibilities to include the protection of children's rights, prevention of child abuse and exploitation, rehabilitation of children requiring care and protection, and welfare support for elderly and disabled persons. Field officers such as **Probation Officers, Social Service Officers, Child Rights Promotion Officers, Development Officers, and other departmental officers** also contribute to identifying beneficiaries and implementing programs at community level.



Figure 4: office of the organization

The Department continues to implement programs aimed at improving social welfare while supporting vulnerable groups to participate more actively in the development of the Southern Province. Its vision focuses on becoming a leading provider of welfare services for elderly, needy, and disabled persons while creating a secure future for children. More information about the history, functions, vision, and mission of the Department is available on the official website: <https://swpcs.sp.gov.lk/dweb/about.html>

1.1.1 Vision of the company

To be a forerunner of welfare services for the elderly, destitute and disabled persons in the Southern Province, while creating a secure tomorrow for children in the Southern Province.

1.1.2 Mission of the company

To implement projects and programs that provide welfare services for elderly, destitute and disabled persons and enable them to become stakeholders in the development of the Southern Province, while protecting children's rights and providing care and proper guidance to children affected by social injustice.



Figure 5:Organizational Structure Chart

WELFARE INVENTORY AND DISTRIBUTION MANAGEMENT SYSTEM (WIDMS)

2 TRAINING AND EXPERIENCE GAINED

2.1 Introduction – General Introduction About the Training

My industrial training was carried out at the **Department of Social Welfare, Probation and Child Care Services – Southern Province**, District Secretariat, Galle, from **25 May 2026 to 30 August 2026**. During the training period, I was assigned an individual software development project titled **Welfare Inventory and Distribution Management System (WIDMS)**.

The main objective of the project was to computerize and improve the existing process of managing welfare beneficiaries, aid requests, approvals, inventory, procurement, goods distribution, returns, and other related activities carried out by the Department. Since this was an individual project, I was responsible for almost the entire software development life cycle, beginning with requirement gathering and continuing through system analysis, database design, user interface design, implementation, testing, debugging, documentation, and system improvement.

The training provided a different experience from an ordinary academic software project because the system was developed for a **real government working environment**. At the beginning, I had limited knowledge of government welfare procedures, administrative responsibilities, approval structures, and the terminology used by officers. Therefore, before developing the system, I had to understand the existing manual processes and how different officers participated in each workflow.

Another important aspect of the training was communicating with users who did not necessarily have technical knowledge of software systems. Instead of asking them technical questions, I had to understand their daily activities, existing documents, problems, responsibilities, and expected outcomes and then translate these into system requirements. This helped me develop practical skills in **requirement elicitation, communication, business process analysis, and user-centred system design**.

The project requirements also changed several times during development as the actual procedures became clearer. Therefore, the system had to be continuously modified and improved. This gave me practical experience in managing changing requirements and understanding that software development in a real organization is an iterative process rather than a fixed sequence of activities.

2.2 Detailed Description of the Training

2.2.1 Detailed Description of the Training

The first stage of my training involved understanding the working environment of the Department of Social Welfare. Since the organization followed government administrative procedures that were new to me, I initially had to understand how welfare assistance was requested, verified, approved, stored, and finally distributed to beneficiaries.

I communicated with the responsible officers and studied the responsibilities of different users, including the **Administrator, Subject Officer, Store Keeper, and Social Service Officer**. I also identified how beneficiary information, stock information, approval documents, welfare items, and distribution records were handled.

One of the main challenges at this stage was that users explained the processes using their own administrative terminology rather than software terminology. Therefore, I had to convert their descriptions into system workflows, database entities, user roles, and functional requirements.

This experience improved my ability to understand an unfamiliar business environment and convert real organizational activities into a computerized information system.

2.2.2. Requirement Gathering an Analysis

Requirement gathering was one of the most important tasks of my project. Discussions were conducted with the responsible officers to identify the problems in the existing process and the features required in the new system.

The major requirements identified included:

- Beneficiary registration and management
- Aid distribution requests
- Beneficiary eligibility checking
- Administrative approvals
- Goods requests
- Inventory and stock management
- Supplier and payment management

- Welfare item distribution
- Return management
- Vision Camp management
- Contact Lens management
- User and role management
- Activity logging and reporting

Because most stakeholders were not software professionals, I could not simply ask them what database tables, modules, or functions were required. Instead, I asked about **what they currently do, which documents they use, who approves each activity, what problems occur, and what information they need to see.**

I then transformed these requirements into system functions.

⇒ **Changing Requirements**

One of the most significant challenges encountered during the project was the **continuous change of requirements.**

For example, some workflows that originally appeared to be a single process later had to be divided into multiple stages. One important change was the separation between an **Aid Distribution Request** and a **Goods Request**. After an Aid Distribution Request is approved, the Subject Officer creates a separate Goods Request, which goes through another approval stage before the Store Keeper releases the goods.

Other changes affected:

- User responsibilities
- Approval stages
- Beneficiary eligibility rules
- Inventory procedures
- Supplier payment handling
- Vision Camp processing
- Contact Lens handling
- Distribution procedures

Whenever a requirement changed, I analyzed which parts of the system were affected and modified the related **database structure, PHP logic, user interface, validation, and workflow**.

This taught me how to manage requirement changes without losing the overall consistency of the system.

2.2.3 Preparation of the Software Requirement Specification

After gathering the initial requirements, I prepared the **Software Requirement Specification (SRS)** document.

The SRS helped organize the identified requirements into functional and non-functional requirements. It also helped define the system users, their responsibilities, major modules, business rules, and workflows.

Although the SRS was prepared during the early stage, it had to be updated because several requirements changed during system development.

This experience helped me understand that an SRS is not only an academic document but also an important reference for controlling and communicating system requirements.

2.2.4 Database Design

I designed the database according to the identified business processes.

The system required more than a simple user and inventory database because different activities needed to be tracked separately. The database therefore contains entities related to:

- Users and roles
- Beneficiaries
- Aid requests
- Goods requests
- Inventory items
- Stock receipts
- Suppliers
- Supplier payments

- Distributions
- Returns
- Vision camps
- Contact lenses
- Geographic information
- Approval activities
- Audit/activity logs

The final database design contains approximately **42 related tables**, which allowed the system to separate different business processes while maintaining relationships between them.

One challenge was changing the database structure when new requirements were introduced. Instead of rebuilding the entire database each time, I used migration-based modifications and updated the affected relationships.

This improved my understanding of **normalization, primary and foreign keys, relationships, constraints, data integrity, and database migration.**

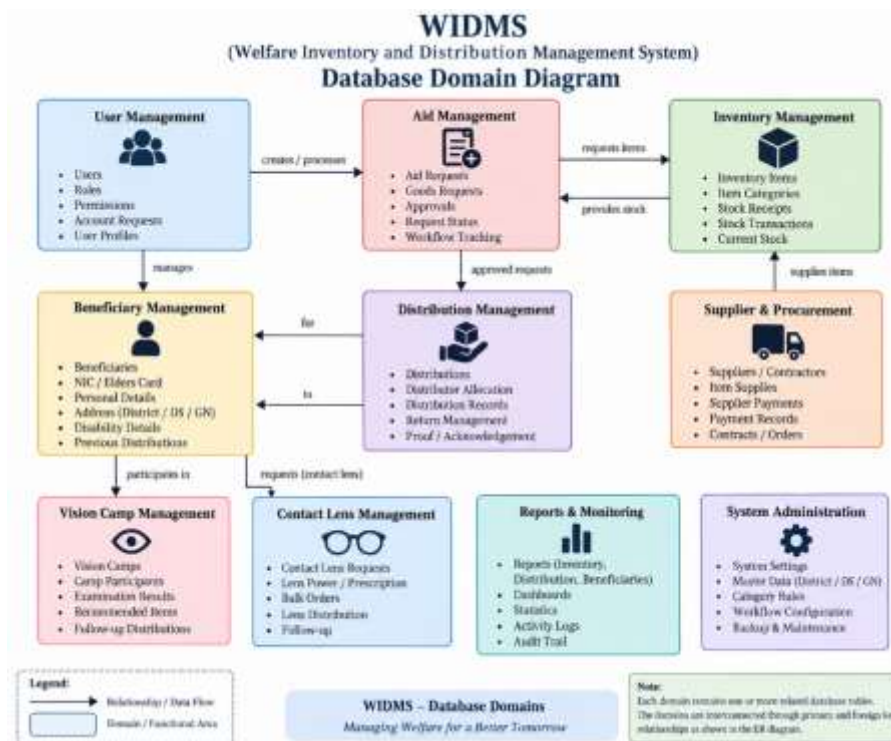


Figure 6 : Database domain diagram

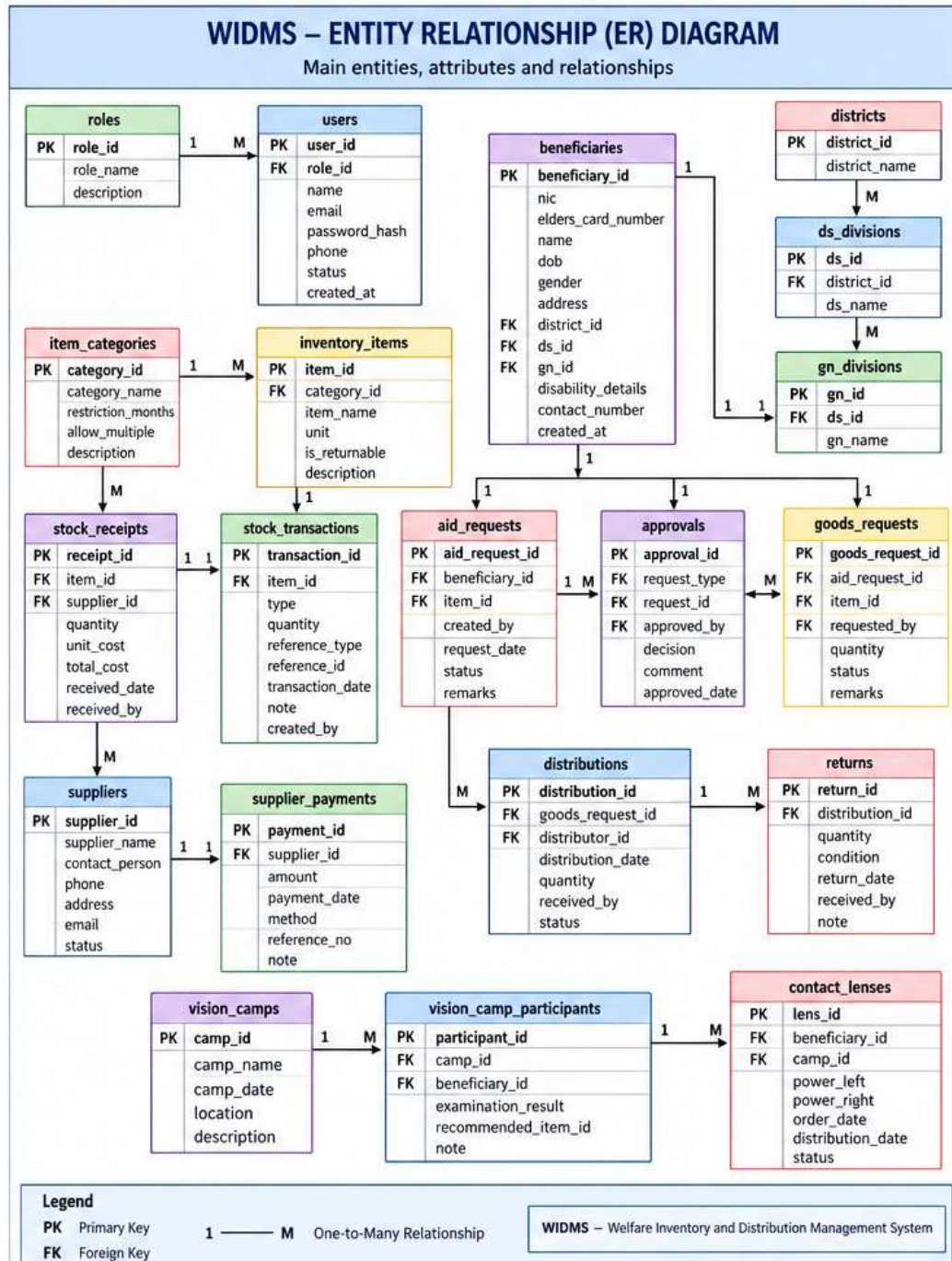


Figure 7:ER Diagram

2.2.5 User Interface and User Experience Design

Another major responsibility was designing the user interfaces.

Since many intended users were government officers who might not have advanced technical knowledge, I tried to keep the interfaces simple and task-oriented.

Separate dashboards were designed for:

- Administrator
- Subject Officer
- Store Keeper
- Social Service Officer

Each user is shown only the functions relevant to their responsibilities.

⇒ A Unique UI Example – New Aid Request Form

One important interface I developed was the New Aid Request form.

Instead of displaying only a normal beneficiary form, I designed it according to the actual administrative process. The interface includes:

- District
- Divisional Secretariat Division
- Grama Niladhari Division
- Beneficiary information
- NIC identification
- Elders' Identity Card identification
- Date of birth
- Gender
- Contact details

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- Address
- Disability information
- Requested welfare item

This form acts as the starting point of the welfare distribution workflow.

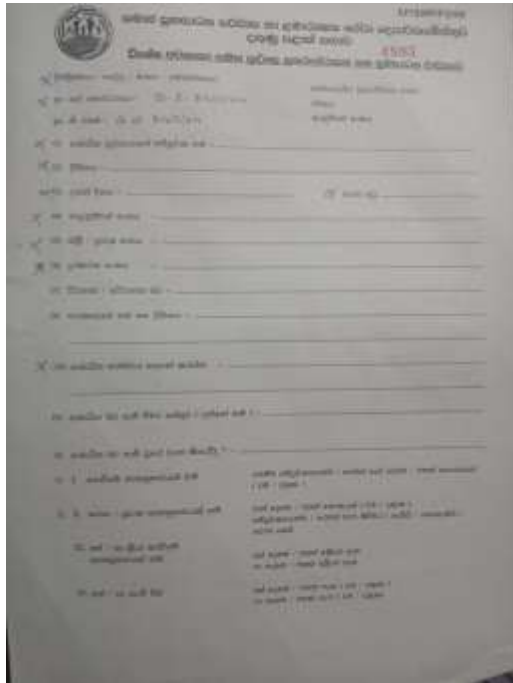
A photograph of the first page of a physical form. It features a header with a logo and text in Hindi. Below the header, there are several sections with labels in Hindi and corresponding input fields for personal and address details. The form is printed on white paper with black text and lines.

Figure 8: page 1 of actual form

A photograph of the second page of a physical form. It contains more sections with labels in Hindi and input fields, including a section for 'Requested welfare item'. There are some handwritten notes and stamps on the page. The form is printed on white paper with black text and lines.

Figure 9: page 2 of actual form

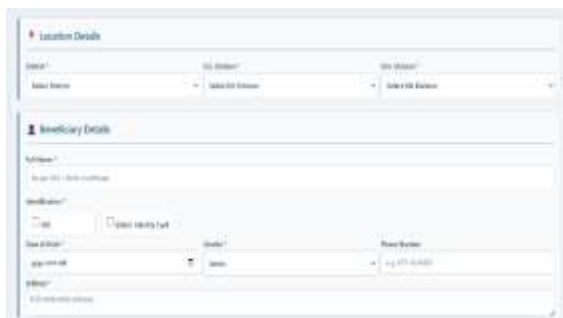
A screenshot of the 'Location Details' section in a web application. It shows a form with fields for 'Area', 'Sub Area', and 'Pin Code'. Below this, there is a 'Secretary Details' section with fields for 'Name', 'Designation', and 'Contact No.'. The interface is clean with a light blue header and white background.

Figure 10: forms upper view in sso ui

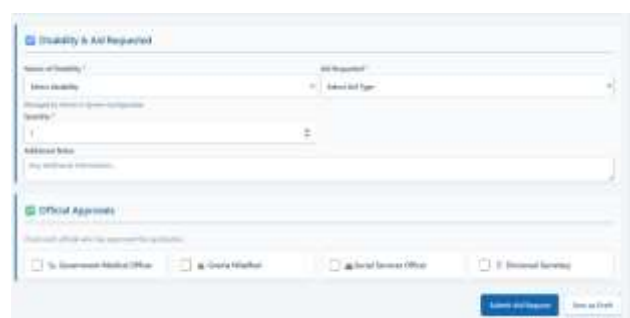
A screenshot of the 'Disability & Aid Requested' section in a web application. It shows a form with fields for 'Disability Type', 'Aid Type', and 'Remarks'. Below this, there is an 'Official Approvals' section with checkboxes for 'To Government Medical Officer', 'To District Medical Officer', 'To District Social Services Officer', and 'To District Secretary'. The interface is clean with a light blue header and white background.

Figure 11: forms down view in sso ui

2.2.6 Dynamic Geographic Selection

Another useful feature was the geographic selection mechanism.

Instead of requiring users to manually type location names, the system follows the hierarchy:

District → DS Division → GN Division

When the user selects a District, the relevant Divisional Secretariat Divisions are loaded. After selecting the DS Division, the corresponding GN Divisions are displayed.

This improves data consistency and reduces spelling errors and incorrect geographic combinations.



```
const district = document.getElementById('district_id');
ds = document.getElementById('ds_division_id');
gn = document.getElementById('gn_division_id');
dob = document.getElementById('date_of_birth');
age = document.getElementById('beneficiary_age');

if (district) {
  let { $ } = return
  document.getElementById('parent_id').value = ''
  const filter = (parent, parent_id) => {
    const { value, options } = parent_id;
    if (value === '') return
    const visible = parent_id.value === '' ? '' : ''
    const hidden = visible
    const disabled = visible
    if (selectedOptions[0].disabled) selected.value = ''
    selected.disabled = parent_id.value
  }

  district.addEventListener('change', () => {
    ds.value = ''
    gn.value = ''
    filter(ds, district.value);
    filter(gn, '');
  });

  ds.addEventListener('change', () => {
    gn.value = ''
    filter(gn, ds.value);
  });

  filter(ds, district.value);
  filter(gn, ds.value);
  const addEventListener('change', () => {
    if (dob.value) {
      const { value } = dob;
      const birth = new Date(value);
      const today = new Date();
      let years = today.getFullYear() - birth.getFullYear();
      if (
        today < new Date(today.getFullYear(), birth.getMonth(), birth.getDate())
      ) {
        years--;
      }
      age.textContent = `Age: ${Math.abs(years)}`;
    }
  });
}
```

Figure 12: code of dynamic geographical selection

2.2.7 Beneficiary Identification

The system supports beneficiary identification using either:

NIC

or

Elders' Identity Card

This was introduced because not every beneficiary may be identified through the same type of document.

The interface dynamically handles the selected identification method while the application validates and stores the relevant information.

This feature demonstrates how the system was adapted to the **actual characteristics of welfare beneficiaries**, instead of forcing all users into a single identification mechanism.

2.2.8 Beneficiary Eligibility Checking

Another important part I implemented was the **beneficiary eligibility checking mechanism**.

Different welfare item categories may have different restrictions. Therefore, eligibility could not be implemented using one fixed rule.

The system considers configurable rules such as:

- Restriction period
- Whether multiple distributions are allowed
- Previous distributions within the category

For restricted categories, the system checks the beneficiary's previous distribution history before allowing another request.

This feature reduces duplicate or inappropriate welfare assistance and supports the Department in following its distribution policies.

2.2.9 Two-Level Request and Approval Workflow

One of the most important workflows developed during the project was the separation of **Aid Requests** and **Goods Requests**.

The process follows approximately:

Social Service Officer → Aid Request → Administrator Approval → Subject Officer → Goods Request → Administrator Approval → Store Keeper → Goods Release

This requirement became clear during later discussions and required changes to the original system design.

Implementing the workflow required modifications to:

- Database tables
- Request statuses
- Approval logic
- Dashboard information
- User permissions
- Store Keeper functions

This was a valuable example of how I responded to a **significant requirement change during development**.

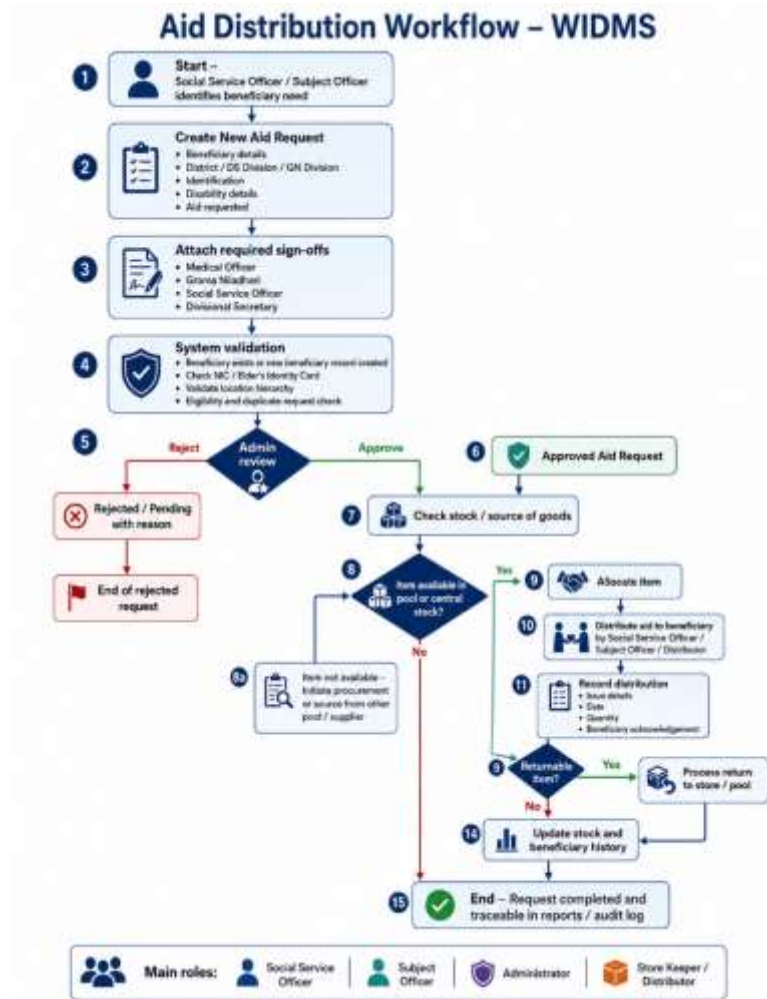


Figure 13: workflow diagram

2.2.10 Role-Based Access Control

Since several officers use the system, each user must only be allowed to access functions related to their responsibilities.

I therefore implemented **role-based access control**.

After login, users are directed to their corresponding dashboards. Server-side checks are also performed to prevent unauthorized access to restricted modules.

For example:

- Administrator manages approvals and system-level functions.

- Subject Officer manages goods-related processes and related activities.
- Store Keeper manages received stock and goods release.
- Social Service Officer manages beneficiaries and Aid Distribution Requests.

This improved both system usability and security.

```
✓ if (!hasRole('social-service-officer')) {  
    http_response_code(403);  
    ⚡ exit('Only Social Service Officers can create aid requests.');
```

Figure

14:role checking

2.2.11 Inventory and Stock Management

The inventory module was designed to maintain records of welfare items received by the Department.

The Store Keeper can record items received together with information such as supplier, quantity, and cost.

The system also maintains records when stock is released for approved requests, helping to track stock movement rather than only maintaining a final quantity.

This provided practical experience in implementing **transaction-oriented inventory management**.

2.2.12 Supplier and Payment Management

Another requirement that appeared during development was managing suppliers and payments.

The system records supplier information, stock receipts, payment records, and partial payments.

This helped me understand that a real inventory management system requires financial and procurement-related records in addition to basic stock quantities.

2.2.13 Special Workflow – Contact Lens Management

Contact Lens distribution required different handling from ordinary welfare items.

The system therefore supports lens-specific information such as the required **lens power** and related bulk-order processing.

This demonstrated that all inventory items cannot always follow exactly the same workflow.

Creating a specialized workflow while keeping it connected to the main system improved my understanding of designing software for **exceptional business cases**.

2.2.14 Special Workflow – Vision Camp Management

The project was later expanded to support **Vision Camp management**.

This module required additional data and processes separate from normal welfare item distribution.

The Vision Camp workflow was therefore implemented as a specialized module while still sharing common beneficiary, inventory, and administrative information with the main system.

This gave me experience in expanding an existing architecture without creating an entirely separate application.

2.2.15 Security and Data Validation

While developing the system, I also had to consider security and data correctness.

Some of the practices applied include:

- Session-based authentication
- Role checking
- Secure password handling
- Prepared database queries
- Server-side validation
- Client-side form validation
- CSRF protection

- Controlled access to system modules
- Activity logging

These mechanisms helped reduce invalid input, unauthorized access, and unsafe database operations.

```
function csrfToken(): string
{
    if (empty($_SESSION['csrf_token'])) {
        $_SESSION['csrf_token'] = bin2hex(random_bytes(32));
    }

    return $_SESSION['csrf_token'];
}

Windsurf: Refactor | Explain | Generate Function Comment | ✕
function verifyCsrfToken(string $token): bool
{
    return isset($_SESSION['csrf_token']) && hash_equals($_SESSION['csrf_token'], $token);
}
```

Figure 15:csrf token

2.2.16 Testing and Debugging

Testing was carried out continuously during development.

Because a change in one workflow could affect another part of the system, I tested individual functions as well as complete workflows.

Testing included:

- Login and authorization
- Account approval
- Beneficiary registration
- Aid requests
- Eligibility validation
- Approvals
- Goods requests
- Stock receiving
- Goods release

- Distribution
- Returns
- Vision Camp functions
- Contact Lens functions

Errors identified during testing were corrected and the affected workflow was tested again.

This taught me that testing should be performed continuously, especially when requirements change frequently.

2.2.17 Working as Multiple Software Development Roles

Since WIDMS was developed as an **individual project**, my responsibilities were not limited to software coding. During the project I performed several roles that would normally be divided among different members of a software development team.

I worked as a:

Requirement Analyst – communicated with officers and identified system requirements.

Business Analyst – studied existing welfare procedures and converted them into system workflows.

System Analyst – identified modules, users, data flows, and business rules.

Database Designer – designed and maintained the relational database.

UI/UX Designer – designed interfaces suitable for government officers.

Backend Developer – implemented server-side business logic using PHP.

Frontend Developer – implemented forms, dashboards, validation, and interactive behavior.

Tester – tested each module and complete system workflows.

Technical Documenter – prepared the SRS, diagrams, database documentation, and project report.

This experience helped me understand how the different stages of software engineering are interconnected and provided broader practical experience than working only on one programming task.

2.2.18 Challenges Encountered and Solutions Applied

Challenges	Action/Solution
Completely new government environment	Learned the existing processes through discussions, observation, documents, and repeated clarification with officers.
Users had limited software/technical knowledge	Asked about their actual daily activities instead of technical system requirements and translated these activities into software functions.
Requirements changed frequently	Followed an iterative development approach and updated the database, UI, business logic, and documentation when necessary.
Complex approval workflow	Modelled the workflow in stages and separated Aid Requests from Goods Request
Different welfare items required different rules	Introduced configurable eligibility rules and specialized workflows.
Geographic data was complex	Implemented District → DS Division → GN Division hierarchical selection.
Multiple user responsibilities	Implemented role-based dashboards and access control.
Database grew as the system expanded	Used structured relational design and migration-based database updates.
Bugs appeared after requirement changes	Re-tested affected modules and complete workflows after each major modification.

These challenges provided some of the most valuable learning experiences during my training. Instead of treating requirement changes and unfamiliar processes as only technical problems, I learned to analyze their causes, communicate with stakeholders, redesign affected components, and validate the solution through testing.

2.2.19 Unique Technical and UI Contributions

This is another section that will make your report look more individual and less generic.

You can state that the main distinctive contributions of your implementation included:

- **District → DS Division → GN Division dynamic geographic filtering**
- **NIC / Elders' Identity Card beneficiary identification**

- **Configurable beneficiary eligibility restrictions**
- **Two-stage Aid Request and Goods Request workflow**
- **Four official sign-off handling**
- **Role-specific dashboards**
- **Contact Lens power-specific workflow**
- **Vision Camp workflow**
- **Stock receipt and goods release tracking**
- **Supplier partial-payment management**
- **Activity/audit tracking**
- **Special workflow statuses that show users the current processing stage**

2.3 Experience Gained

During the industrial training period, I gained practical experience that extended beyond programming. Developing WIDMS as an individual project allowed me to experience the complete software development life cycle and understand the relationship between organizational requirements and technical implementation.

The project improved my knowledge of **PHP, MySQL/MariaDB, HTML, CSS, JavaScript, Bootstrap, relational database design, authentication, validation, role-based access control, debugging, testing, and documentation**. More importantly, I gained experience in communicating with non-technical stakeholders, understanding unfamiliar administrative processes, managing frequently changing requirements, and converting real-world procedures into a structured software system.

Working in a government environment also helped me understand that successful software development depends not only on technical knowledge but also on communication, patience, requirement analysis, adaptability, and continuous stakeholder feedback. The experience gained through WIDMS has therefore improved both my technical skills and my ability to approach real-world software engineering problems systematically.

3. OBSERVATIONS AND COMMENTS

3.1 Observations

During the training period, I observed that developing an information system for a government organization requires greater attention to **accuracy, accountability, traceability and administrative procedure**. A technically correct solution is not sufficient if it does not reflect the responsibilities and authorization levels followed by the organization.

Another important observation was the value of maintaining **structured and consistent data**. Information such as beneficiary details, administrative locations, welfare items, requests, approvals and distributions must be recorded accurately because these records may be referred to later for decision-making and reporting. Dropdown selections, validation rules, predefined statuses and database relationships were therefore useful for improving data consistency.

I also observed that each system user requires a different level of information. Administrative users need an overall view of requests, approvals and system activities, while operational users require interfaces related specifically to their daily responsibilities. Providing role-specific functions makes the system easier to understand and reduces unnecessary access to unrelated operations.

The project also demonstrated the importance of maintaining historical information. Welfare distribution is not limited to recording the current stock balance. Previous assistance received by a beneficiary, request history, approval history, stock movements and return records are also important for future decisions.

Another observation was that exceptional processes should not always be forced into a general workflow. Certain welfare activities contain additional information or processing requirements and therefore benefit from separate modules while still sharing common system data.

Overall, the training helped me understand that a successful information system must balance **technical implementation with organizational requirements and practical usability**.

3.2 Evaluation of the Developed System

The developed WIDMS provides a centralized approach to managing welfare-related information. Beneficiary records, requests, approvals, inventory information, distributions and related activities can be maintained within a single system instead of being handled through disconnected records.

The use of role-based access provides better separation of responsibilities. Validation mechanisms also help reduce incorrect or incomplete data entry. Maintaining request statuses and activity records improves the visibility of the progress of each process.

The relational database structure makes it possible to connect information between different modules while avoiding unnecessary duplication. This is particularly useful when reviewing beneficiary history, inventory movements and completed distributions.

From a usability perspective, the system was designed with structured forms, clear navigation, status indicators, search facilities and role-specific dashboards. These features are intended to make the application easier to use in an operational environment.

Therefore, the current implementation can be considered a useful foundation for improving the management and monitoring of welfare inventory and distribution activities.

3.3 Current Limitations

Although the major functions of WIDMS have been implemented, the system would require further preparation before full operational deployment.

The application has mainly been developed and tested in a local development environment. A production implementation would require a secure server environment, HTTPS configuration, regular database backups, recovery procedures and appropriate system monitoring.

Additional testing with actual users from each role would also be beneficial. User Acceptance Testing could identify usability improvements that may not be visible during developer testing.

Geographical master data may require future updates when official administrative information changes. Therefore, such updates should be performed through a controlled process to protect existing records.

The current active-request restriction is applied according to the beneficiary and aid category. If future organizational policy requires a stricter rule covering all aid categories, this validation can be extended.

3.4 Recommendations

Several improvements can be considered for future development of WIDMS.

The system should be deployed on a secure production server with **HTTPS, automated database backups and recovery procedures**. Formal User Acceptance Testing should be conducted before organization-wide use.

Notification facilities could be introduced to inform responsible officers about requests that require attention or approval. Additional reporting and data visualization features could also improve management-level monitoring.

Future versions could provide more advanced search and filtering, automated report generation, improved accessibility and further multilingual support. A controlled mechanism for maintaining master data such as administrative divisions and welfare categories would also improve long-term maintainability.

The system should continue to be updated according to organizational policies and operational feedback.

4. CONCLUSION

The industrial training provided an opportunity to apply academic knowledge to a practical software development environment through the development of the **Welfare Inventory and Distribution Management System (WIDMS)**.

The completed system demonstrates how a web-based information system can support the management of welfare beneficiaries, inventory and distribution-related activities while maintaining structured records and administrative control. The project also provided valuable exposure to the complete development process required to transform an organizational requirement into a working software solution.

Through this training, I gained a better understanding of the importance of analysing real organizational procedures before designing a system. I also improved my ability to connect database design, application logic, user interfaces, validation and security mechanisms into a complete application.

In addition to improving my technical knowledge, the training strengthened my communication, problem-solving, documentation, time-management and decision-making skills. It also provided practical experience in working independently and taking responsibility for the development of a real-world system.

Overall, the industrial training was a valuable learning experience that allowed me to connect theoretical knowledge with practical software engineering. The experience and knowledge gained during the development of WIDMS will be useful in my future academic work and professional career.

REFERENCES

- [1] Department of Social Welfare, Probation and Child Care Services – Southern Province, “Official Website.” Available:
<https://swpcs.sp.gov.lk/>
- [2] Department of Social Welfare, Probation and Child Care Services – Southern Province, “About the Department.” Available:
<https://swpcs.sp.gov.lk/dweb/about.html>
- [3] Department of Social Welfare, Probation and Child Care Services – Southern Province, “Social Welfare Staff.” Available:
<https://swpcs.sp.gov.lk/dweb/staff2.php>
- [4] District Secretariat – Galle, “Official Website.” Available:
<https://eapps.galle.dist.gov.lk/>
- [5] PHP Documentation, “PHP Manual.” Available:
<https://www.php.net/manual/en/>
- [6] MySQL, “MySQL Documentation.” Available:
<https://dev.mysql.com/doc/>
- [7] Bootstrap, “Bootstrap Documentation.” Available:
<https://getbootstrap.com/docs/>
- [8] MDN Web Docs, “JavaScript Documentation.” Available:
<https://developer.mozilla.org/en-US/docs/Web/JavaScript>
- [9] OWASP Foundation, “Web Application Security Resources.” Available:
<https://owasp.org/>
- [10] Composer, “Dependency Manager for PHP.” Available:
<https://getcomposer.org/>
- [11] PHPMailer, “PHPMailer Documentation.” Available:
<https://github.com/PHPMailer/PHPMailer>

APPENDICES

Appendix A – Project Development Timeline

Include the summarized timeline from **25 May 2026 to 30 August 2026**, showing the major stages of the project such as requirements gathering, SRS preparation, database design, implementation, testing and documentation.

Appendix B – System Architecture and Database Design

Include:

Figure B.1 – WIDMS System Architecture

Figure B.2 – WIDMS Database Domain Diagram

Figure B.3 – WIDMS Entity Relationship Diagram

Appendix C – Main Workflow Diagrams

Include:

Figure C.1 – Aid Distribution Workflow

Figure C.2 – Goods Request and Stock Release Workflow

Figure C.3 – Procurement and Stock Workflow

Figure C.4 – Vision Camp Workflow

Figure C.5 – Contact Lens Workflow

Appendix D – Selected User Interfaces

Include only the most important screenshots:

Figure D.1 – Login Interface

Figure D.2 – Officer Account Request Interface

Figure D.3 – Administrator Dashboard

Figure D.4 – Social Service Officer Dashboard

Figure D.5 – New Aid Request Form

Figure D.6 – Subject Officer Dashboard

Figure D.7 – Store Keeper Receive Items Interface

Appendix E – Existing Manual Documents

Include the two photographs of the existing welfare assistance form.

Figure E.1 – Existing Manual Welfare Assistance Form – Beneficiary Information

Figure E.2 – Existing Manual Welfare Assistance Form – Recommendations and Authorizations

These documents can be included as evidence of the existing procedure studied during the project.

Appendix F – Selected Important Code

Include only short code extracts for:

- Dynamic District → DS Division → GN Division selection
- Beneficiary eligibility checking
- Duplicate active request checking
- Aid Request creation
- Role-based access control
- CSRF protection

Appendix G – Testing Summary

Include a small testing table showing major functions such as login, access control, Aid Request submission, location validation, eligibility checking, approval processing, inventory updates and return handling with their **expected result and actual result**.

Appendix H – Project File Structure

Include the main WIDMS directory structure showing:

```
config/  
database/  
includes/  
modules/  
public/  
scripts/  
vendor/
```

This appendix can demonstrate how the project source code was organized into configuration, database, reusable components, role-based modules and public application files.